

Mohammad Tawhidul Hasan Bhuiyan

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RESEARCH INTEREST

Compilers, Hardware/Software Co-Design

EDUCATION

Columbia University

PhD in Electrical Engineering

New York, USA

Sept. 2024 – May 2030 (expected)

Bangladesh University of Engineering and Technology (BUET)

Masters of Science in Computer Science and Engineering

Palashi, Dhaka, Bangladesh

July 2021 – Aug. 2024

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Computer Science and Engineering

Palashi, Dhaka, Bangladesh

Mar. 2016 – Feb. 2021

🎓 *Class rank: 1st*

EMPLOYMENT

Columbia University

Graduate Research and Teaching Assistant

New York, USA

Sept. 2024 – Present

Bangladesh University of Engineering and Technology (BUET)

Lecturer, Department of CSE

Dhaka, Bangladesh

Sept. 2021 – Aug. 2024

PUBLICATIONS

- **Tawhid Bhuiyan**, Sumya Hoque, Angelica Moreira, Tanvir Ahmed Khan, **Wax: Optimizing Data Center Applications With Stale Profile**, *ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS), Volume 2, Association for Computing Machinery (ACM)*, 2026
- Ryan Piersma, **Tawhid Bhuiyan**, Tanvir Ahmed Khan, and Simha Sethumadhavan, **Reverse Engineering DVFS Mechanisms**, *2025 IEEE International Symposium on Hardware Oriented Security and Trust (HOST), San Jose, CA, USA, 2025*
- **Mohammad Tawhidul Hasan Bhuiyan**, Muhammad Rashed Alam and M. Sohel Rahman, **Computing the Largest Common Almost-Increasing Subsequence**, *Theoretical Computer Science, Volume 930, 2022, Pages 157-178, ISSN 0304-3975*
- **Mohammad Tawhidul Hasan Bhuiyan**, Irtesam Mahmud Khan, Sheikh Saifur Rahman Jony, Renee Robinson, Uyen-Sa D. T. Nguyen, David Keellings, M. Sohel Rahman, and Ubydul Haque, **The Disproportionate Impact of COVID-19 among Undocumented Immigrants and Racial Minorities in the US**, *International Journal of Environmental Research and Public Health, Volume 18(23): 12708, 2021, PubMedID 34886437, ISSN 1660-4601*

INVITED TALKS

- Feb. 5, 2026: Presented *WAX: Optimizing Data Center Applications With Stale Profile* to Meta
- Mar. 9, 2026: Presented *WAX: Optimizing Data Center Applications With Stale Profile* to Google

TECHNICAL SKILLS

Architectural Simulators: Accel-sim, Astra-sim

Compiler & Systems: LLVM, BOLT

Performance Tools: perf, NVIDIA Nsight, NVML, CUPTI, PyTorch Profiler

Languages: C/C++, Python, Java, SQL, Bash, JavaScript

Libraries: Polars, Numba, PyTorch, Scikit-Learn, Tensorflow, Pandas, NumPy, Matplotlib

AWARDS

- BUET Alumni Association Award for securing top position in the department
- Dean's Honor List, BUET in all 8 undergraduate semesters
- University Merit Scholarship, BUET

TEACHING & MENTORSHIP

Columbia University

Mentor

- Mentored Amir Zarandi (Undergraduate student, Columbia University) in conducting original research, resulting in a co-authored paper which is under submission.
- Mentored 15 students across 4 technical projects. Guided teams through the design, implementation, and debugging complex systems. Some key projects include:
 - *Fine-grained GPU DVFS*: Leading the students to update Accel-sim to allow simulating and evaluating new DVFS techniques for GPU cores.
 - *HBM DVFS & Co-Optimization*: Leading the investigation into HBM idle power; utilizing DRAMsim3 and Ramulator2 to propose co-optimized DVFS mechanisms for GPU compute and memory.
 - *Power-Aware Collective Communication*: Guiding the integration of different ML models into Astra-sim to easily simulate different model architecture.

Bangladesh University of Engineering and Technology (BUET)

Lecturer

- Developed curriculum and delivered lectures for core undergraduate courses:
 - *Operating Systems*: Led hands-on sessions on modifying open-source kernels; taught memory virtualization techniques and file systems.
 - *Database Systems*: Instructed on schema design, normalization theory, and the internal architecture of DB storage systems.

REFERENCES

Tanvir Ahmed Khan

Assistant Professor

Department of Electrical Engineering

Columbia University

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